

## Worksheet 6: Floral Formula & Family Solanaceae

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

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| <b>Student Name:</b> _____ | <b>Roll No.:</b> _____ | <b>Section:</b> _____ |
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### Section A: Multiple Choice Questions (Select the single best answer)

- The floral formula of Family Solanaceae exhibits which of the following gynoecium characteristics? [ \_\_\_\_ ]**
  - Monocarpellary, ovary inferior with marginal placentation
  - Bicarpellary syncarpous, superior ovary with axile placentation and swollen placenta
  - Tricarpellary syncarpous, superior ovary with parietal placentation
  - Polycarpellary apocarpous, inferior ovary with basal placentation
- A persistent calyx that remains attached even after fruit formation is a diagnostic feature of: [ \_\_\_\_ ]**
  - Mustard (Brassicaceae)
  - Pea (Fabaceae)
  - Brinjal / Tomato (Solanaceae)
  - Lily (Liliaceae)
- Which pair of plants belonging to Family Solanaceae are cultivated for their potent medicinal properties? [ \_\_\_\_ ]**
  - Nicotiana tabacum and Petunia
  - Atropa belladonna and Withania somnifera (Ashwagandha)
  - Solanum tuberosum and Capsicum annuum
  - Solanum melongena and Solanum lycopersicum
- In the floral formula of Solanaceae, the epipetalous stamen condition is represented by: [ \_\_\_\_ ]**
  - A subscript (5) under K
  - An arc connecting the symbols C(5) and A5
  - A bar placed beneath G
  - A percentage (%) sign preceding the formula

### Section B: Floral Formula Interpretation & Technical Symbols

- Write the complete semi-technical floral formula for Family Solanaceae and define what each symbol signifies.**

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- State the meaning of the following floral symbols: (a) %, (b) Br, (c) G, (d) P.**

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### Section C: Economic Importance Classification Table

| Economic Category | Common Plant Examples   | Botanical Names |
|-------------------|-------------------------|-----------------|
| Food & Vegetables | Potato, Tomato, Brinjal | _____<br>—      |
| Spices            | Chilli                  | _____<br>—      |

|            |                         |       |
|------------|-------------------------|-------|
| Medicinal  | Belladonna, Ashwagandha | _____ |
|            |                         | —     |
| Fumigatory | Tobacco                 | _____ |
|            |                         | —     |
| Ornamental | Petunia                 | _____ |
|            |                         | —     |

### Section D: Diagnostic & Descriptive Questions

7. Summarize the key vegetative and floral diagnostic features of Family Solanaceae with respect to leaf venation, calyx aestivation, androecium attachment, and fruit types.

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Lecturer Signature: \_\_\_\_\_